

README

05/21/2025

1. Architecture of the folders

code

This folder contains the codes for reproducing the analysis in Sections 5, 6 and the supplementary material of the paper (simulation studies and real data studies).

- **simul.R**: All results of simulation in the main manuscript (Section 5) can be reproduced by this file.
- **simul_supp.R**: All results of simulation in the supplementary materials (Section S9) can be reproduced by this file.
- **simul_CV.R**: Tuning parameters for simulation in the main manuscript (Section 5) can be selected by this file.
- **data_analysis.R**: All results of real data analysis in the main manuscript (Section 6) and the supplementary materials (Section S8) can be reproduced by this file.

There is also a sub-folder named “functions” which contains the codes for functions to implement our proposed methods and others.

- **spcovreg.R**: functions to implement our proposed method.
- **spcovreg_basic.R**: some simple functions such as the soft thresholding operator which have been used in `spcovreg.R`, `simul.R`, `simul_supp.R`, `simul_CV.R` and `data_analysis.R`.
- **covreg_em.R**: functions to implement the covariance regression by Hoff and Niu (2012).

data

This folder contains three sub-folders for all the utilized data and output from both simulation and real data analyses.

- **main**: All data and output of simulation (Section 5) and real data analysis (Section 6) in the main manuscript are included in this folder. There are three types of RData files.
 - **CVresult_n200p50q30_MA_set1.RData**: This is the pregenerated list of tuning parameters (α and λ) selected by running the code “`simul_CV.R`”. This file is loaded when running the code “`simul.R`” (see the code). Other files for different settings (e.g. $n = 500$ and $q = 100$) can be obtained by running “`simul_CV.R`” after modifying the code.
 - **fit_n200p50q30_MA_set1.RData**: This is the pregenerated set of results by running the code “`simul.R`”. This file is loaded in Section 2 of this README file to produce numbers and figures. Other files for different settings (e.g. $n = 500$ and $q = 100$) can be obtained by running “`simul.R`” after modifying the code.
 - **gbmdata_run.RData**: This is the pregenerated set of results by running the code “`data_analysis.R`”. This file is loaded in Section 2 of this README file to produce figures.

- **supp**: All the utilized data and output of simulation in the supplementary materials (Section S9) are included in this folder. There is one RData file in this folder.
 - **supp_q1.RData**: This is the pregenerated set of results by running the code “simul_supp.R”. This file is loaded in Section 2 of this README file to produce numbers and figures.
- **GBMdata**: Raw data files for 178 patients with glioblastoma multiforme (GBM) from the REM-BRANDT study (GSE108476). There are 6 raw files in this folder. The code for preprocessing these data can be found in “data_analysis.R”.
 - **clinical.data.csv**: This data include age and gender data for the patients with GBM or other diseases. We considered only the GBM patients.
 - **gene_names.csv**: This data contain the name of the genes considered.
 - **expData_73.RData**: This data contain the gene expression for the 73 genes considered.
 - **final_id.RData**: data for patient ID.
 - **SNPname.RData**: list of names for the 118 SNPs considered.
 - **SNPdata4.RData**: data for the 118 SNPs considered.

2. Reproducing the results in the manuscript and the supplementary materials

```
##### set working directory #####
# please specify the path
path="/Users/rakheon_kim/Desktop/Research/SparseCovReg/"
```

Numbers for tables: MA(1) model under Setting 1 with $n=200$, $p=50$, $q=30$

Numbers for other settings can be obtained by loading the R data for each setting.

```
load(file=paste0(path,"data/main/fit_n200p50q30_MA_set1.RData"))
# Table 1
mean(S_diff); mean(ST_diff); mean(covreg_diff); mean(lse_diff); mean(covlpd_diff)
```

```
## [1] 3.219506
```

```
## [1] 2.014333
```

```
## [1] 7.810432
```

```
## [1] 15.08602
```

```
## [1] 1.874898
```

```
sd(S_diff); sd(ST_diff); sd(covreg_diff); sd(lse_diff); sd(covlpd_diff)
```

```
## [1] 0.07939699
```

```
## [1] 0.06482416
```

```
## [1] 0.3448368
```

```
## [1] 0.3379816
```

```
## [1] 0.06224962
```

```
# Table 2
```

```
mean(sqrt(covl_err_est)); sd(sqrt(covl_err_est)); mean(lasso_TPR); mean(lasso_FPR)
```

```
## [1] 4.581958
```

```
## [1] 0.1651217
```

```
## [1] 0.7662121
```

```
## [1] 0.007572249
```

```
# Table 3
```

```
colMeans(tb_COV)
```

```
## [1] 0.9380469 0.9269388 0.9380612 0.9502101 0.9475510 0.9502136
```

Figure 1 (Section 5)

```
load(file=paste0(path,"data/main/fit_n500p50q30_MA_set1.RData"))
```

```
# Code for creating the picture can be found in simul.R
```

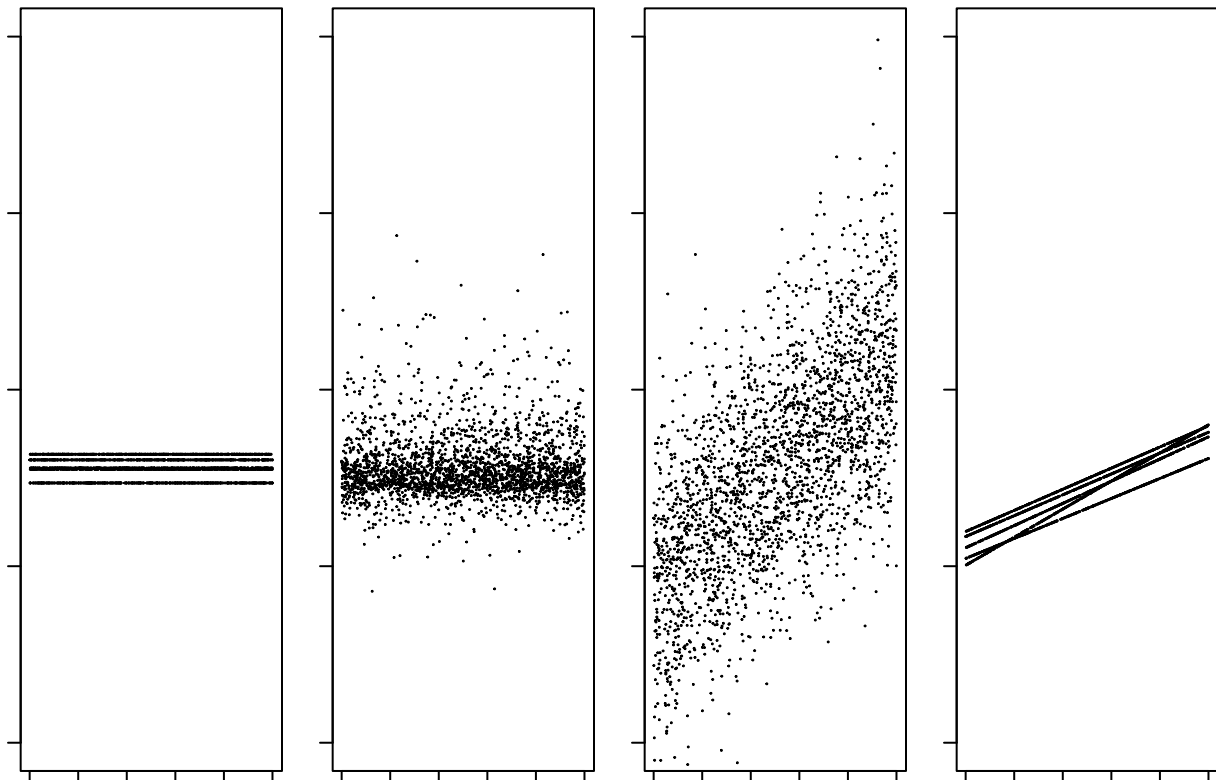


Figure 2 (Section 5)

```
load(file=paste0(path,"data/main/fit_n500p50q100_MA_set2.RData"))  
# Code for creating the picture can be found in simul.R
```

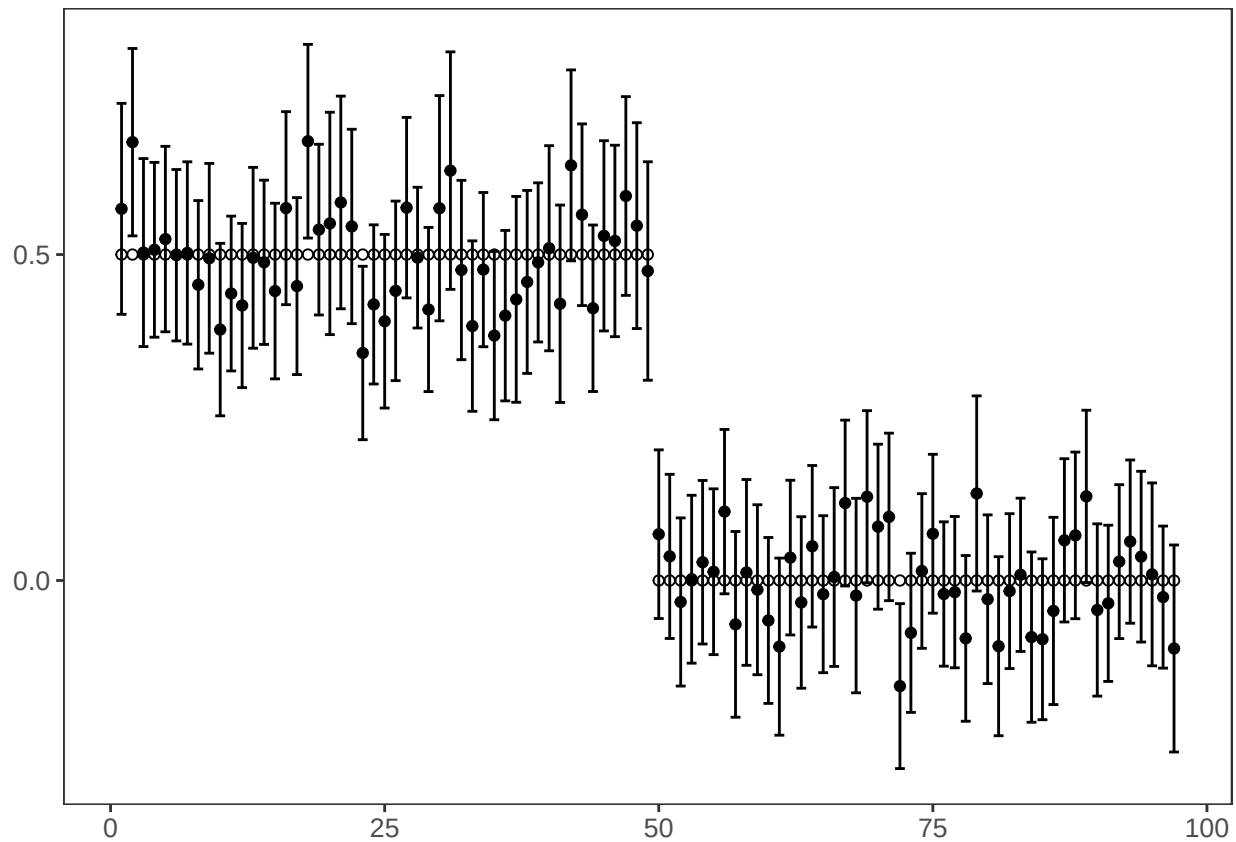


Figure 3 (Section 6)

```
load(file=paste0(path,"data/main/gbmdata_run.RData"))  
library(reshape2)  
# Code for creating the picture can be found in data_analysis.R
```

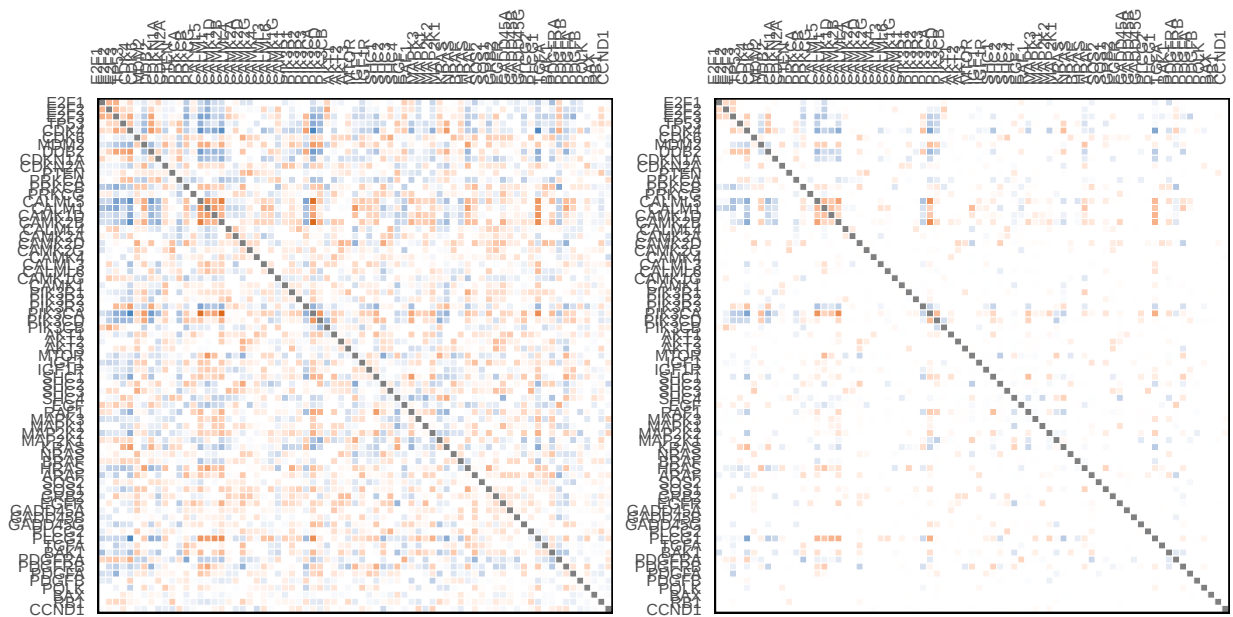


Figure 4 (Section 6)

Code for creating the picture can be found in data_analysis.R

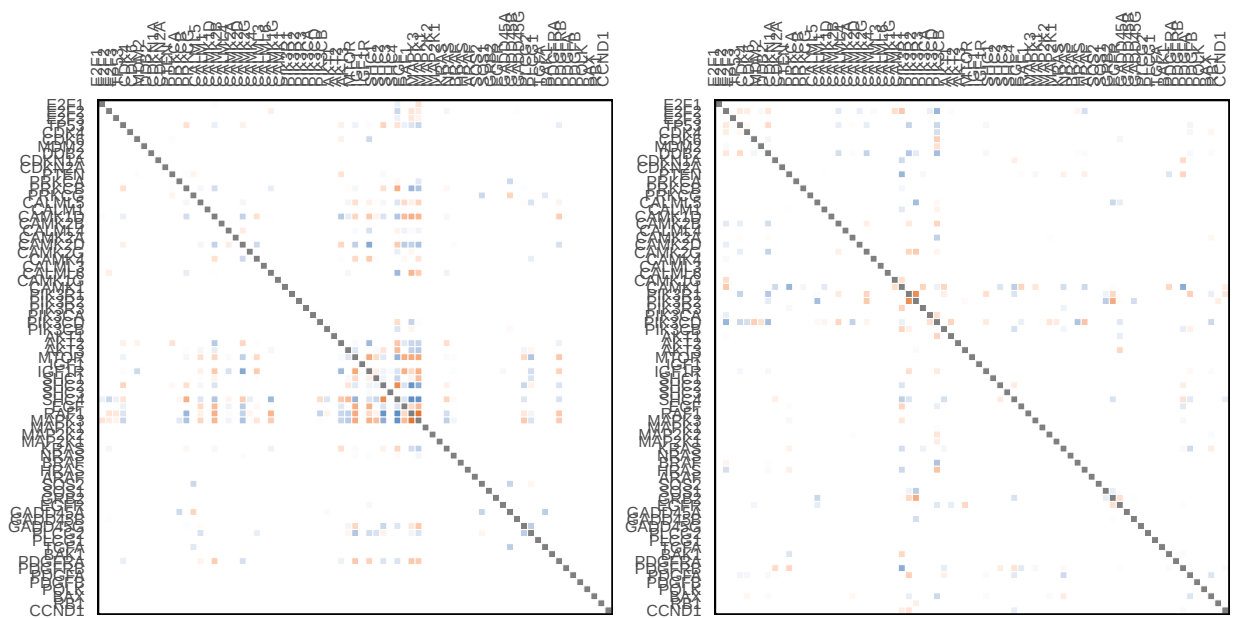
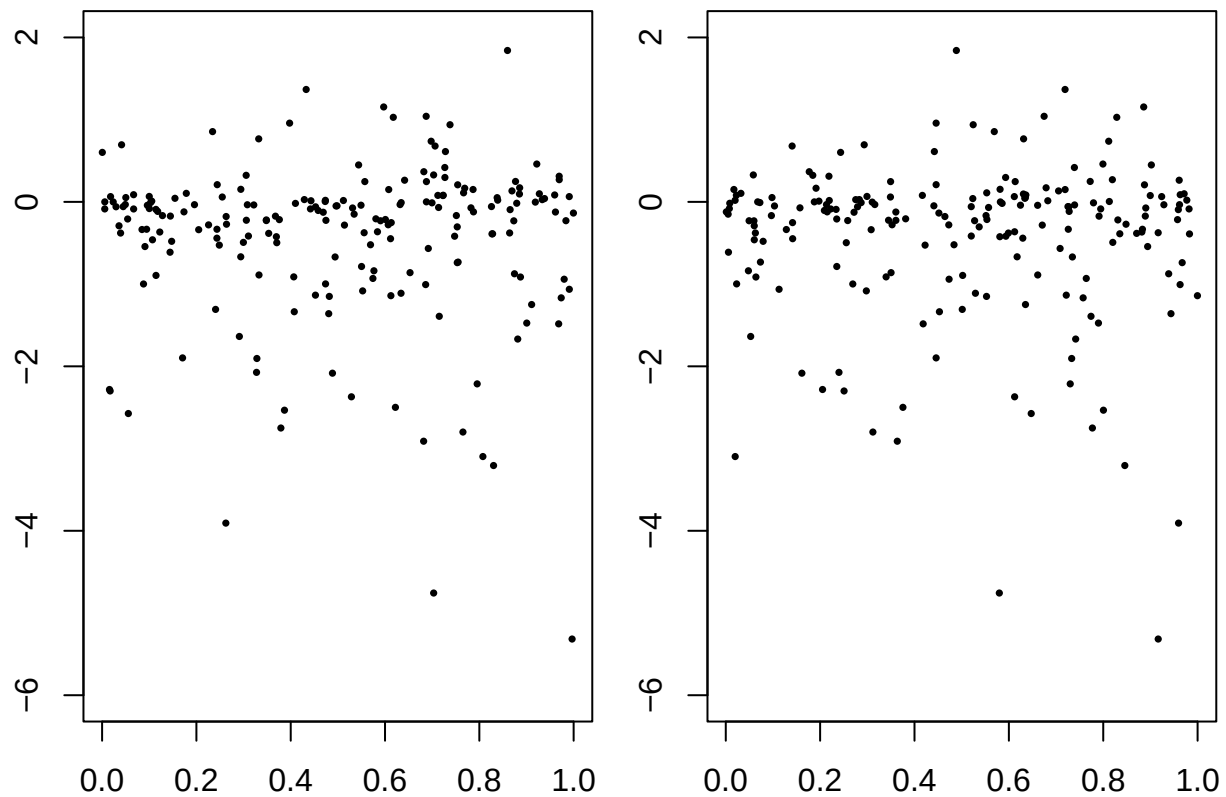


Figure 5 (Section 6)

`library(network)`

`##`



Numbers for tables in Section S9 (when $q=1$)

Numbers for other settings can be obtained by loading the R data for each setting.

```
load(file=paste0(path,"data/supp/supp_q1.RData"))
# Table S1
mean(covreg_mcmc_diff); mean(covl_diff3); mean(covl_diff)
```

```
## [1] 0.2308865
```

```
## [1] 0.2518296
```

```
## [1] 0.2898408
```

```
sd(covreg_mcmc_diff); sd(covl_diff3); sd(covl_diff)
```

```
## [1] 0.09205443
```

```
## [1] 0.08425484
```

```
## [1] 0.0943065
```

All of these can be obtained <https://github.com/rakheon/SparseCovReg>.